

Code Explanation of Course-Management-System

1. Relation between classes in UML class diagram

Admin to student, instructor, and courses

Admin is responsible of managing students, for recording, preserving their data, and activities. And every student is independent and admin just control them. Also is similar activity for instructors too. Admin has access to all courses and record them, and there is relation for reporting registering and managing them. The relation between admin and other classes is aggregation.

Instructor and courses

Each instructor can create several courses and managing it. Course and instructor have direct relations. Also, each courses have one instructor. This is composition relation.

Student and courses

Courses can have many records by many students also many students can register in many courses. This relation to enrollment is aggregation, but the relation to review and rating is dependence. Additionally, student send feedback or rate a course, then a dependence relation created between course and student.

Recommendation and courses

Recommendation is dependent class to student, for producing recommend.

GUI and Admin

GUI is dependent class, that needs admin information and features to produce the graphical inter face. Makes aggregation relationship.

2. Detailed description for each method and attribute

Student class attribute

Student ID: makes that there is unique identification in the database, it is also important for user authentication in the platform.

Name: shows personal identity in interface, for increasing user experience and better identification.

Email: for the needs of any communication and notifications, account recovery, and supporting platform involvement.

Registered modules: tracks students current and past registration. Also, monitoring progress and preventing duplicate registration are two of main purpose.

Preferences: store student interests, also is key concept for personalized recommendations.

Register date: is for users that help for analysis loyalty, sending notifications, and prioritization.

Student class methods:

generate ID: this method generates random id but uniquely for specific user.

Preferences: return interests of user,

Module register: this method handles course registration by verifying factors and updating them. it integrates with payment section if needed.

Course review: this allows student to send their feedback, which relates the course's ratings and informs future recommendations. Improves course qualities through related reviews.

Is registered: checking for registration, practical for prerequisite and record idea.

Course class attribute:

Course ID: identification for courses as a unique for each, to referencing in registrations, searches, and recommendations.

Title: this represents course's name it helps to browsing easily and making decision.

Description: shows detail insights into course content, it also helps student to access easily.

Instructor ID: make a relation between course and its instructor, enable instructor to manage and create course.

Price: represent the price of course that is useful for identifying popularity.

Category: this classifies courses by their subjects, it helps for filtering and subject-based recommendations.

Rating: collecting student feedback, which affect course popularity and recommendation algorithms.

Review: represent reviews and record assessment by student for the course

Enrolled student, create date: are the registered list of students and the date in the course.

Course class methods:

Generate ID: this method generates random id but uniquely for specific course.

Create course: by this, instructors can upload and organize learning materials, being up to date. It is also, important for course management and for content and instructor control.

Managing: it computes the average rating from all reviews, providing a summary score for display and ranking. Helps student to evaluate courses, also used in recommendation scoring.

Add rating: handling scores, that is useful in identifying popularity for recommendation.

Add review: represents recording ideas textual

Register student: adding new student to the list.

Recommendation class attributes:

Student info: contains student information like: preferences, history of registration, and performance. It is for analysing user and trends.

Course view: it includes all course details such as category, rating, and content. Helps to be sure for recommendation planning.

Recommendation class method:

Gen recommendation: represent generate recommendations, for its algorithm, to suggest courses. It accesses student info and course view to personalize suggestions. Increasing user involvement for relevant courses and increasing platform preservation.

Calculate scores: for handling scores such as preferences quality and all scores that effect on recommendation.

Instructor class attributes

Instructor ID: identifying instructors by id also for courses and managing accounts

Name: for illustrating in profile and courses screen

Email: for the needs of any communication and notifications, account recovery, and supporting platform involvement.

Expertise: identifying skills to help classification and searching.

Manage courses: creating a list of courses that is provided by instructor

Feedback: the ideas for recording about classes and courses

Instructor class methods

Generate instructor id: this method generates random id but uniquely for specific instructor.

Create course: adding new courses, necessary for registration

Manage course: managing course like edit, remove, add, and publish.

Admin class attributes

Students: this is a dictionary that collect all students in system

Instructors: this is a dictionary that collect all instructors in system

Courses: this is a dictionary that collect all courses in system

Admin key: for authentication of admin

Activity log or Activities: a list of activities register, courses, payment and other necessary factors.

Admin class methods

Register user, manage enrolment, and review submission, admin have access to add instructor or students, also publishing, editing all sent content.

GUI

Graphical User Interface designed class-based, for this educational system for three groups, of instructors, students and admins to have suitable interact with system. This class have some methods that has some tabs for students, instructors, courses, enrollment and report.

Setter and Getter

These are two types of object-oriented-program methods to controlled access to each feature of class. So, by this class features become readable and changeable.

3. Comprehensive strategy for generating personalized course recommendations

To ensuring accuracy and relevance recommendation strategy is provided, also combined multiple data sources. This strategy uses user activity, course content and relation reviews.

Data collecting and processing

User Activity: record user activities like course registration, time spent on lessons, exam scores, and search history. This data builds clear preference profile and information.

Course Content: features inside courses, including category, keywords from descriptions, difficulty level, and instructor specialization.

Relations Reviews: benefits of ratings and reviews to assess course quality.

User profiling

Creating dynamic profile for each student.

Clear preferences: from filling surveys or manual setting like multiple choice selection.

Implicit preferences: dedicated from activity patterns, such as frequency in access to checking courses or high performance in math exams.

Behavioral patterns: comes from registration history and completion rates to identify learning goals.

Recommendation techniques

Filtering on content: the similarity of courses for those students who have engaged with can be recommended. Use a brief similarity on courses features like category and key word. For example, if a student rated (Python intermediate) course highly, suggest (Advanced Python) based on content.

Collaborative filtering: Identify those who have similar profiles using K-nearest neighbor (KNN) algorithm. Recommend those courses that have been selected by similar students, rated highly, but for those who have not taken it yet. This makes strong behavior for discovery.

Hybrid approach: combining content filtering scores to make a balance personalize and diversity. For example, measuring subject related matches with interests and collaborative signals from users for popular courses.

Incorporating related reviews: to weight recommendations, course rating will be used. The first prioritize is for those courses with high rating on behalf of student in similar profiles. For example, those courses with good score for (clear explanations), if the student values teaching style.

User performance adjustment: recommending to user based on their academic performance. For example, if a student is higher than beginner courses, suggest intermediate, or if they are not in that level recommend prerequisites or materials.

Real-Time and contextual factors

Recommendations should be updated dynamically in this level based on recent activity. For example, after completing a course, suggest higher course or follow up the course.

Considering contextual data that could be device type or the time of day, but in educational concepts or relevance.

Evaluate and optimization

Measurements the important and effectiveness factors is in this level. Clicking rates, registration rates, registration satisfaction after all (after a course has been completed).

Use tests to refine algorithms, to ensuring recommendations are related in platform aims of increasing engagement and learning outcomes.

Summary

This strategy ensures that recommendations are exactly fitted with user learning paths, utilizing a lot of sources of data to suggest courses, and feedback. Also, the system learns a personalized learning environment, enhancing student willing, and preserving the education platform.

Also, this explanation has a source code.